Kurapati Srinivasa Abhiram

Indian Institute of Science Education and Research, Pune | Indian Institute of Technology Hyderabad

abhiram.srinivasa@gmail.com https://in.linkedin.com/in/srinivasa-abhiram line-id: ks.abhiram

CAREER OBJECTIVE

I am a Material Scientist. I have good experience in testing the thermal and electrical properties of semiconducting materials. I got my master's thesis with a distinction. Now I am working as an TEEP intern in Taiwan at NCUE, Changhua with emphasis on designing and developing thermoelectric devices.
I am very much interested in working in Foxconn's semiconductor team to contribute to the R&D or process engineering. I believe my experience in lab work and data analysis would be a great asset in turning materials into devices, my career objectives strongly overlaps with the $3+3+3$ framework of Foxconn, for the vision of advancing Semiconductor field.

EDUCATION

Indian Institute of Technology Hyderabad (IIT-HYD) **Hyderabad, India**
MS Thesis, Department of Materials Science and Metallurgical Engineering *May 25–Jun 26*
Ranked 7th in India, 588 in the QS World University Rankings 2027.

Indian Institute of Science Education and Research Pune (IISER) **Pune, India**
BS–MS Dual Degree, Physics with Distinction in Masters Thesis Project *June* 2026
Ranked 101-150 in the 2020 (Times Higher Education) THE Young University Rankings.

RESEARCH PROJECTS

[1] Efficient Synthesis of Bismuth-Based Semiconductors for Functional Applications *Aug 2026– present*
Novel Materials and Thermeoelctrics laboratory, NCUE, Taiwan ***Prof. Chia Ji Liu***
Extensively working on Bismuth – Tellurium based Semiconducting materials for Thermoelectric Device level application.

[2-A] Exploring III-VI Semiconductors for Functional Applications *Dec 2025– June 2026*
Laboratory for Energy and Advanced Devices (LEAD), IIT Hyderabad ***Dr. Suresh Perumal***
Synthesised III-VI semiconducting materials and studies their structural, optical, and thermal characterization performed. Utilisation of these samples for functional applications are currently being explored.

[2-B] Band Engineering of I-III-VI_2 for Thermoelectric Power Generation *May 2025–Apr 2026*
Synthesised Cu-Sb-Se based chalcogenides via solid-state synthesis and densification through hot pressing. Te alloying was explored to tune thermoelectric properties; in-depth characterization was performed across structural, optical, chemical, and thermoelectric properties.

[3] Encapsulation of Metal Oxides in CNTs *Jan – April 2025*
Exp. Condensed Matter Physics, Nanomagnets & Spintronics, IISERP ***Dr. Ashna Bajpai***
Designing of CNT's via CVD (single and multi-walled) and encapsulated metal oxides; sample property analysis with AC susceptibility measurements and Raman spectroscopy.

[4-A] Improvement of zT in p-type Chalcogenide Material *Apr –Dec 2024*
Quantum Material Growth and Property Measurement Lab, IISERP ***Prof. Surjeet Singh***
Synthesis and zT measurements via Co-doping at the Sb and Te sites with Mn, Zn and Se respectively, in Ag-Sb-Se/Te-based Chalcogenide semiconducting materials through solid-state synthesis.

[4-B] n-type Thermoelectric Half-Heusler Material *Apr – Dec 2024*

Explored n-type TaFeSb Half-Heusler compounds via arc melting and powder synthesis; phase purity was identified as a key challenge, with concurrent analysis of antisite disorder effects on zT in p-type HH compounds.

[5] Experimental Techniques in High Energy Physics *Jan – Apr 2024*
Experimental High Energy Physics Group, IISERP ***Prof. Seema Sharma***
Hands-on experience in electronics through tabletop experiments with detectors, with emphasis on calibration and data acquisition methods. Studied high-energy physics applications in medical imaging.

[6] Analysis of Bacterial Contamination in Establishments *Aug 2022–Dec 2022*
Bugs & Drugs Laboratory, IISERP ***Dr. Nishad Matange***
Investigated microbial colonization patterns and strains, highlighting the importance of hygiene practices.

Other Projects

Development of an open-source LaTeX (TikZ) schematic repository for experimental physics [🐱]
The Community Project: Developing web applications to streamline research workflows, especially TE [🌐]

PUBLICATIONS

Srinivasa Abhiram, Suresh Perumal. Band Engineering of I-V-VI_2 chalcogenides for Thermoelectric Power Generation (In preparation).

CONFERENCES & WORKSHOPS

SPARC Workshop on Advances in Correlative Microscopy and Spectroscopy, IIT H	Feb 2026
Symposium on Magnetism and Spintronics, IIT Hyderabad 🌐	Jan 2026
79th IIM ATM, IIT Hyderabad 🏆 *Best Poster Award*	Dec 2025
Rietveld Refinement of X-Ray Diffraction (RRD), CACR Delhi	Nov 2025
International Online Conference on Crystals (IOCC 2024)	Sep 2024
Sokendai Asian Winter School	Jan 2023
Introduction to Quantum Computing, QKRISHI	Nov 2022

AFFILIATION & PROFESSIONAL EXPERIENCE

Member, Indian Institute of Metals	2025–Present
Member, APS (62168980)	2023–Present
NASA's Future Innovations in Gamma Rays (FIG SAG)	2024–Present
Science Trust Project	2024–Present
Student Ambassador, APS (American Physical Society)	2023–2025

EXPERIMENTAL AND TRANSFERABLE SKILLS

Proficient in a range of synthesis techniques and characterization workflow, with Strong system-level debugger with proven ability to troubleshoot equipment/process issues and translate analytical data into improvements.

Bulk & Nano synthesis | Raman Spectroscopy | UV-Vis-Spectroscopy | EDS/EDX | DSC & TGA | XPS | Electrochemical Workstation | ZEM-3/LSR | BET | LFA | Micro & Nano-Indentation | Glovebox Operation | Arc-Melting | LaTeX | CVD | Induction & Resistive Hot Press | Screen Printing | GitHub

VOLUNTEERING

Thesis Mentoring – B.Tech Student Apr 2026

Co-guided a B.Tech thesis student on the effect of Cd doping in GaSb. Providing support in experimental synthesis and analysis, data interpretation, and result optimization.

Teaching Assistant – Energy Materials and Devices (MS5120) Jan 2026

Designed and developed teaching resources across select course modules.

Volunteer – Poster Committee Nov 2025

Contributed in the planning and organising of a large-scale poster event comprising ∼900 posters.

RELEVANT COURSES

Physics

Classical Mechanics I, II	Electrodynamics	Electronics I, II
Statistical Mechanics	Condensed Matter Physics	Physics Lab I–V
Particle Physics	Energy Materials and Devices	Non-Linear Dynamics
Quantum Mechanics I–IV	Optics	Physics at Nano-Scales
Mathematical Methods	High Energy Physics	Numerical Analysis

Software

- Origin Pro
- CASA XPS
- Crystal Diffract
- VESTA
- X'Pert HighScore
- Thermo-Calc
- ImageJ
- Full Prof
- GABI (LCA)
- Blender
- MS Office
- KiCad
- COMSOL
- Ansys

Courses/ Certifications

Introductory Chinese pronunciation and daily conversation — Sep 2026 (enrolled)

Density Functional Theory, Coursera — Sep 2026 (enrolled)

DSC & TGA E-Learning Courses, NETZSCH — Feb 2026

Introduction to Machine Learning For Engineering Research (ML4ER), UW Madison, USA — Jan 2025

LANGUAGES

- English (C1)
- Hindi (C1)
- Italiano (principiante)
- Telugu (Native)

HOBBIES

- Web Design
- UI-UX
- Cooking
- Photography

You can find an e-copy of this resume at https://shorturl.at/khC1Z